

Thorium - the safe way to energy security

When I was growing up in the 60s many science writers would like to make predictions of the science and technologies of the future - typically for the year 2000, an obvious choice. Looking back, very little of this proved accurate. Take the laser as an example; nobody saw that coming, and even when the invention occurred, nobody had any idea of the extent to which lasers would later be used. So, I am not going to try. But there is one prediction regarding future technologies that I am confident to make - whatever we develop, we are going to need plentiful energy.

The lack of investment in the west in viable energy is very worrying, particularly after the recent problems in Japan, which have heightened irrational fears about nuclear power generation. Many western governments play around with so-called renewable energy sources, such as those ridiculous bird-killing wind turbines. Expensive and next to useless. We need energy generation with balls, not tree-hugging hippy nonsense like this. Ultimately, there is only one truly renewable energy source - the Sun. We may well one day be able to harvest solar energy with sufficient efficiency, but in the meantime, we should go nuclear, big time; not with the well-known technologies, but with thorium.

There is not enough space here to go into the details of how thorium is used in a fission reactor, but in brief: it would usually be used in a liquid salt form, transforming into a uranium isotope after neutron bombardment. Subsequent reactions create further uranium, and the heat given out is turned into electricity.

So, why thorium? There are many reasons. There is much more thorium in the earth than uranium, and India is sitting on something like 20 per cent of

the world's deposits. A neutron beam drives the process and there is no chain reaction; this means no possibility of a melt-down or nuclear explosion. The waste that is created - about one tenth of a per cent of the waste from current reactors - is far less dangerous than from present-day reactors and is removed continuously; you do not have to shut down the reactor to remove spent fuel rods. Remarkably, a thorium reactor can also consume the plutonium and other waste products from normal reactors, thereby helping solve the main problem those reactors create. The thorium process also does not create plutonium as a by-product, and so it cannot readily be used to

create weapons. That latter point speaks volumes for the intentions of governments in continuing with normal uranium-based nuclear generation.

While the west sits on its hands on this one, at least there are some moves in India towards thorium energy generation. These moves should be accelerated strongly, particularly considering how much thorium there is here. China is ahead, and a few months ago announced a thorium reactor development program. My hope is that India and China driving this development will make the west wake up and get moving.

One point that is often not discussed is that electricity may well be a clean form of energy, but it is not always the most convenient, mainly because it is relatively difficult to store. Although we will one day run out of natural oil, I still think hydro-carbons should play a big part in our energy future. It is difficult to imagine a more convenient energy source than hydro-carbons for cars and planes. Electric car - what a silly idea! Also cooking - what a backward step it would be to turn from gas hobs to electric. No real cook would want such a curse.

So, we need to move towards generating hydro-carbon fuels from carbon dioxide and water. There was some research and development in this using catalytic methods, but it proved too expensive at the time. I suggest that given cheap and abundant electrical energy from thorium, it would be well worth our while pouring large sums into developing hydro-carbon fuels. If the time ever comes when fusion reaction is actually made to work, or solar energy capture becomes practical, both of those would produce electricity, and we then convert that into whatever hydro-carbon forms are most convenient. In the meantime, we need thorium. 



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